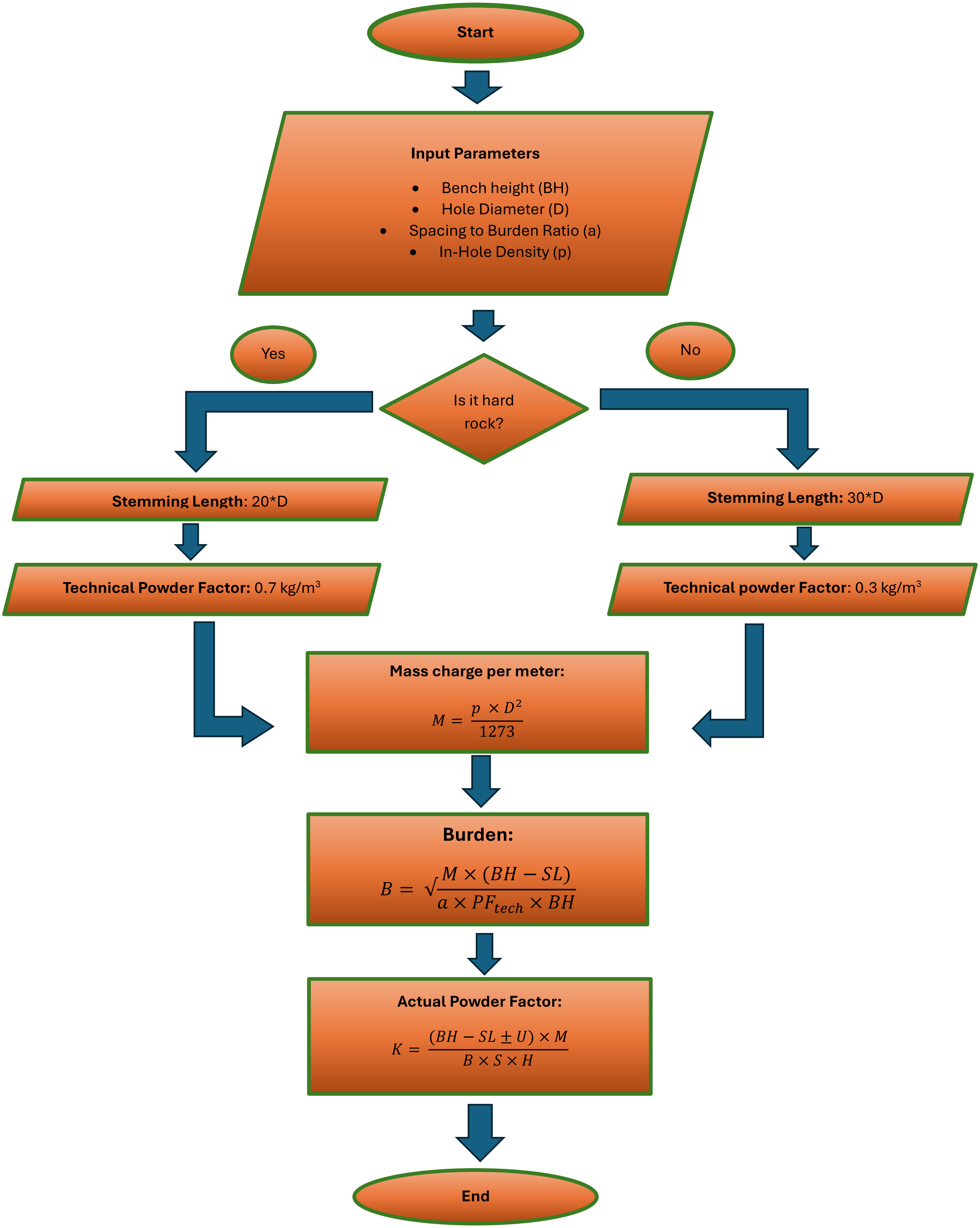
Surface Blast Design Calculator

1. Ask the user to provide the onsite data that contain the blasting specifications which are Blasting Diameter, Average In-Hole Density, Spacing to Burden Ratio and Bench Height.
2. Conditions and restrictions will be set so that accurate and precise values are entered by the blasting engineers.
3. With restrictions and conditions that regard the correct units, the app must display a comment/ instruction that tells the blasting engineer the correct units that variable must have.
4. The first restriction will be such that the Bench height value must be entered in meters (m) and that the app only accepts positive numbers that are greater and not equal to zero.
5. The second restriction will be that the blasting diameter (D) must be in millimetres (mm) and the app only accepts positive numbers that are greater than zero.
6. The third restriction will be that the average in-hole density must be in grams per cubic centimetre (g/cc) and have a positive value that is greater than zero.
7. The app will ask the engineer to specify the type of rock on that environment.
8. The application will then execute a calculation to calculate the stemming length (SL), (using the assumption/ rule that stemming length on hard rock is 20 times hole diameter and in soft rock it is 30 times hole diameter).
9. Still evaluating the type of rock, the user will then be provided a value of the technical powder factor taken from a table of constants that shows the average technical powder factors to be used in each type of rock.
10. Using the average hole density then the app will calculate the mass charge per meter (M) which will be in (kg/m), using the formular [M=(p*D^2)/1273].
11. The app will now calculate burden (B) using the provided parameters and the ones calculated: Bench height, Stemming Length, Spacing to burden ratio, Bench height, Mass charger per meter, Technical Powder factor
12. The application will now use the calculated value for burden to calculate the actual powder factor together with all the other parameters [K= (B-SL+/- U) *M/(B*S*H)].



Start
Input Parameters
Bench height (BH)
Hole Diameter (D)
Spacing to Burden Ratio (a)
In-Hole Density (p)
Yes
Is it hard rock?
No
Stemming Length: 20*D
Stemming Length: 30*D
Technical Powder Factor: 0.7 kg/m3
Technical powder Factor: 0.3 kg/m3
Mass charge per meter:
$M = \frac{p \times D^2}{1273}$
Burden:
$B = \sqrt{\frac{M \times (BH - SL)}{a \times PF_{tech} \times BH}}$
Actual Powder Factor:
$K = \frac{(BH - SL \pm U) \times M}{B \times S \times H}$
End

START

Display “Blast Design Calculator”

INPUT bench height (BH)

INPUT hole diameter (D)

INPUT spacing to burden ratio (a)

INPUT in hole density (p)

IF bench height is less than or equal to zero OR hole diameter is less than or equal to zero OR density is less than or equal to zero OR ratio is less than or equal to zero:

THEN DISPLAY “Invalid input. All values must be positive and greater than zero”

END IF

DISPLAY bench height, hole diameter, ratio and density

INPUT rock type

CONVERT rock type to UPPERCASE

Hard rock constant = 0.7 kg/m^3

Soft rock constant = 0.3 kg/m^3

IF rock type is HARD:

THEN Technical powder factor = hard rock constant And Stemming length = 20 * diameter

Else:

Technical powder factor = soft rock constant And stemming length = 30 * diameter

END IF

CALCULATE mass charge per meter: $M = \frac{p \times D^2}{1273}$

DISPLAY mass charge per meter

CALCULTE burden: $B = \sqrt{\frac{M \times (BH - SL)}{a \times PF_{tech} \times BH}}$

DISPLAY burden

CALCULATE subdrill(U): (1/2 * burden)

CALCULATE spacing(S): (ratio * burden)

CALCULATE charge length (H): (bench height – stemming length)

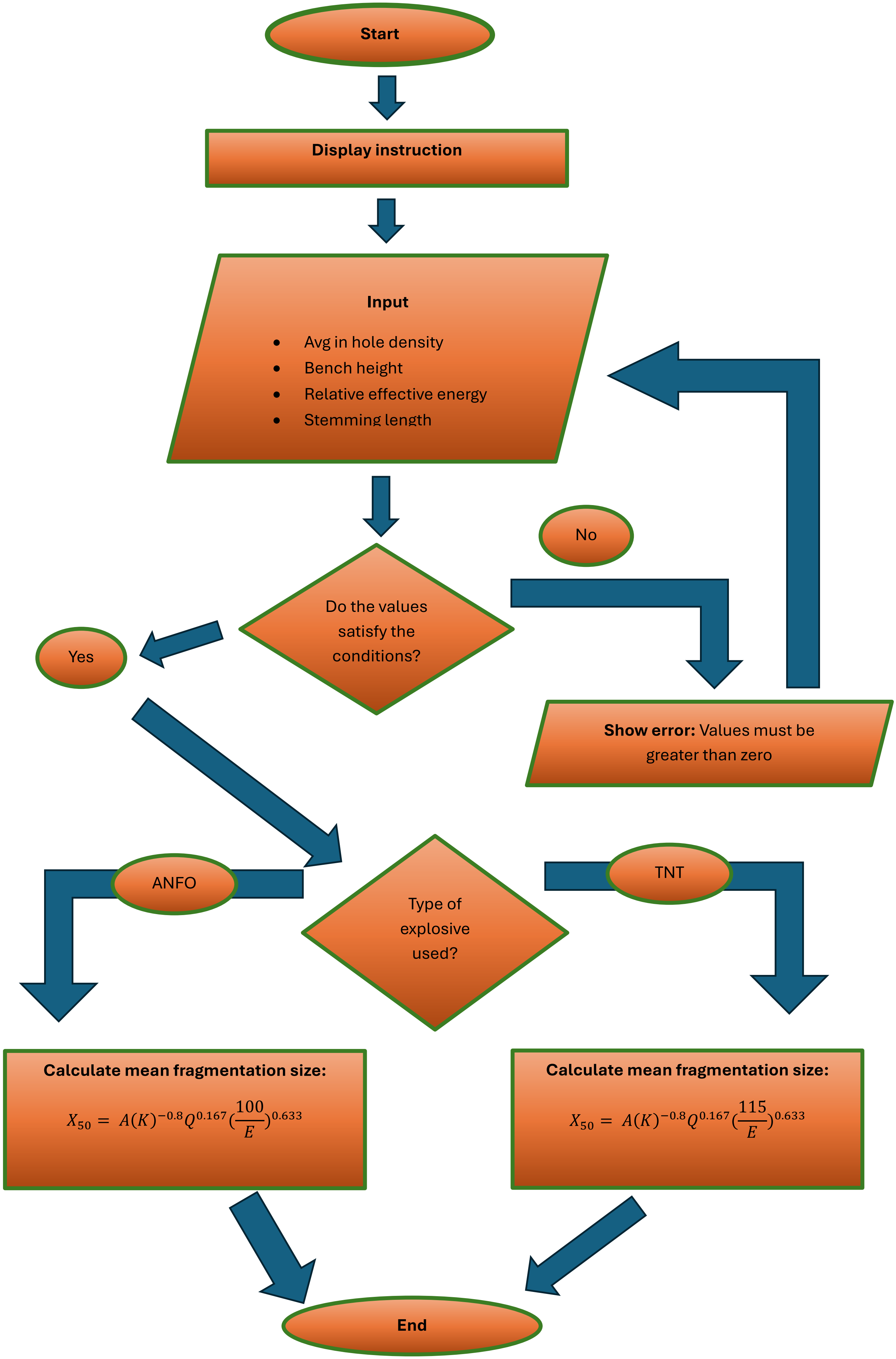
CALCULATE actual powder factor: $K = \frac{(BH - SL \pm U) \times U}{B \times S \times H}$

DISPLAY actual powder factor

END

1. The user is prompted to provide the following input parameters: Average In-hole Density (kg/m^3), Bench Height (m), Relative Effective Energy of Explosive (E) (ANFO = 100, TNT = 115), and Stemming Length (m)
2. The application displays instructions indicating the correct units for each parameter before the user enters the values.
3. The application validates all user inputs by ensuring: All values are numeric, No negative or zero values are accepted, if an invalid value is entered, an error message is displayed, and the user is prompted to enter the value again.
4. Once all valid inputs have been received, the application uses the fragmentation size equation to calculate the mean fragment size (X_{50}).
5. The application displays the calculated mean fragment size in the appropriate unit (e.g., millimeters (mm)).
6. The user is given the option to perform another calculation or exit the application.

```
START
  Display "FRAGMENTATION SIZE CALCULATOR"
  Display "Please enter the following values."
  Display "Average In-hole Density (kg/m³)"
  Display "Bench Height (m)"
  Display "Relative Effective Energy (ANFO = 100, TNT = 115)"
  Display "Stemming Length (m)"
    Input AverageInHoleDensity
  UNTIL AverageInHoleDensity > 0
    Input BenchHeight
  UNTIL BenchHeight > 0
    Input RelativeEffectiveEnergy
  UNTIL RelativeEffectiveEnergy > 0
    Input StemmingLength
  UNTIL StemmingLength > 0
Calculate MeanFragmentSize
 (Use the fragmentation size formula)
  Display "Mean Fragment Size = ", MeanFragmentSize, " mm"
END
```



Start
Display instruction
Input
Avg in hole density
Bench height
Relative effective energy
Stemming length
Do the values satisfy the conditions?
No
Yes
Show error: Values must be greater than zero
Type of explosive used?
ANFO
TNT
Calculate mean fragmentation size:
$X_{50} = A(K)^{-0.8}Q^{0.167}(\frac{100}{E})^{0.633}$
Calculate mean fragmentation size:
$X_{50} = A(K)^{-0.8}Q^{0.167}(\frac{115}{E})^{0.633}$
End